Rutgers Health Hack Challenge 2

Tele-Neurology Operations Command Center: From Visibility to Intervention

Sevaro | Detect coverage risk, recommend an intervention, and act on it

The Challenge

Sevaro provides tele-neurology services to more than 200 hospitals. During a shift, each physician juggles incoming and outgoing hospital calls, video consultations, imaging review and documentation, often for several consults at once. Demand is uneven: a single hospital can generate several stroke consults in half an hour, and that surge can coincide with physicians reaching the end of their shift.

Seeing that this is happening is not enough. Operations and clinical leadership need to act while it still matters: reassign a stranded consult, escalate a delay, or bring an off-shift physician online for a few hours to cover a gap. Today that decision depends on someone noticing, working out who is credentialed and eligible, and reaching them manually.

> **A baseline live view already exists. Do not rebuild it.** Sevaro already operates a live workload view showing physician status, per-physician call and consult counts, and stage timers for stroke consults. A solution that only displays current state will score poorly. The value of this challenge is what happens **after** a problem is detected.

Recreated sketch of the baseline live view Sevaro already runs. This is the starting point, not the target. Values are synthetic.

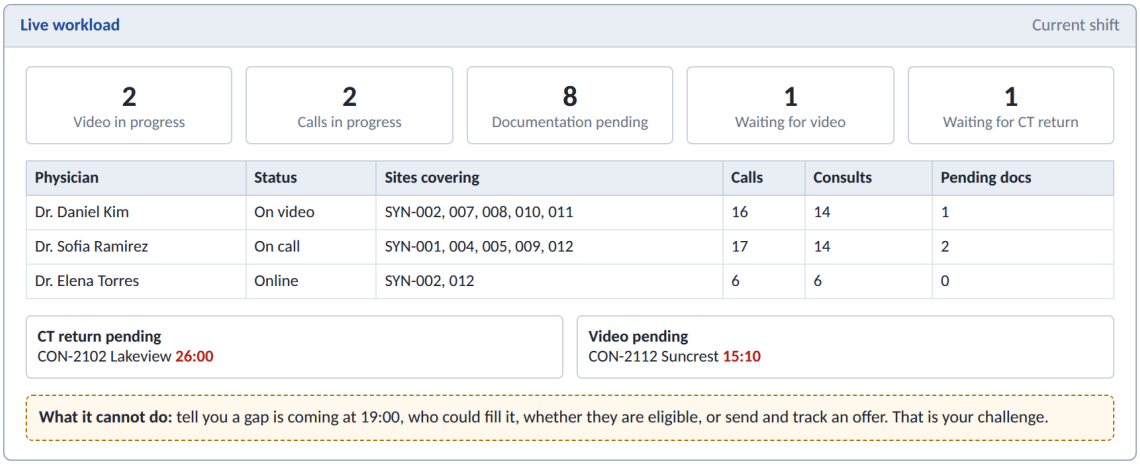


Figure 1. Recreated sketch of the existing baseline view, populated with this challenge's data at 18:20. Build beyond this, not a copy of it.

Problem to Solve

Build an operational system with three connected layers. All three are required.

Layer	Question it answers	Examples
1. Detect	What needs attention now, and what will need attention in the next 60-90 minutes?	Consults breaching time standards; demand above a hospital's baseline; credentialed capacity below the required level now or after upcoming shift ends; physician signals that contradict each other.
2. Decide	What is the right intervention, and who has to approve it?	Reassign a consult; escalate to clinical leadership; offer a split shift. Rank eligible off-shift physicians and explain why each one is or is not eligible.
3. Act	Execute the intervention and close the loop.	Send the offer through an approval step; track accept/decline/timeout; update projected coverage when someone accepts; stop sending offers once the gap is filled.

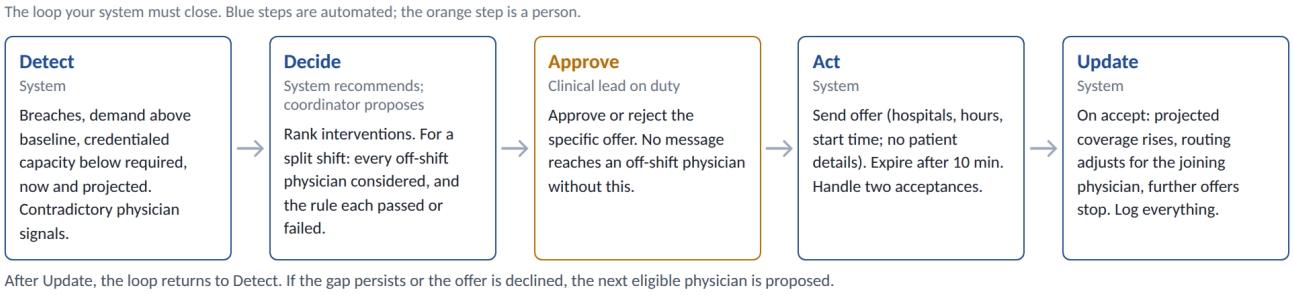


Figure 2. The closed loop. Detect, Decide and Act are required; the approval step is a person.

Users and Decision Rights

User	Needs to see	Can do
Operations coordinator	Network-wide queue, breaches, physician activity, projected coverage gaps	Reassign consults, contact physicians, **propose** split-shift offers
Clinical lead on duty	Gaps that put stroke care at risk; proposed interventions with rationale	**Approve or reject** split-shift offers; authorize escalations
Off-shift physician	A clear offer: hospitals, hours, start time, why they were asked	Accept or decline from a phone; nothing else

Automated actions are permitted only for low-risk steps (for example, an internal alert to the coordinator). Any message to a physician outside their shift requires a human approval in the demo.

Important Data Model Concepts

Physician state and consult state are separate

A physician can have several open consults at once: on a call for Consult A while Consult B waits for CT/CTA and Consult C has a note pending. Represent physician activity and consult lifecycle independently.

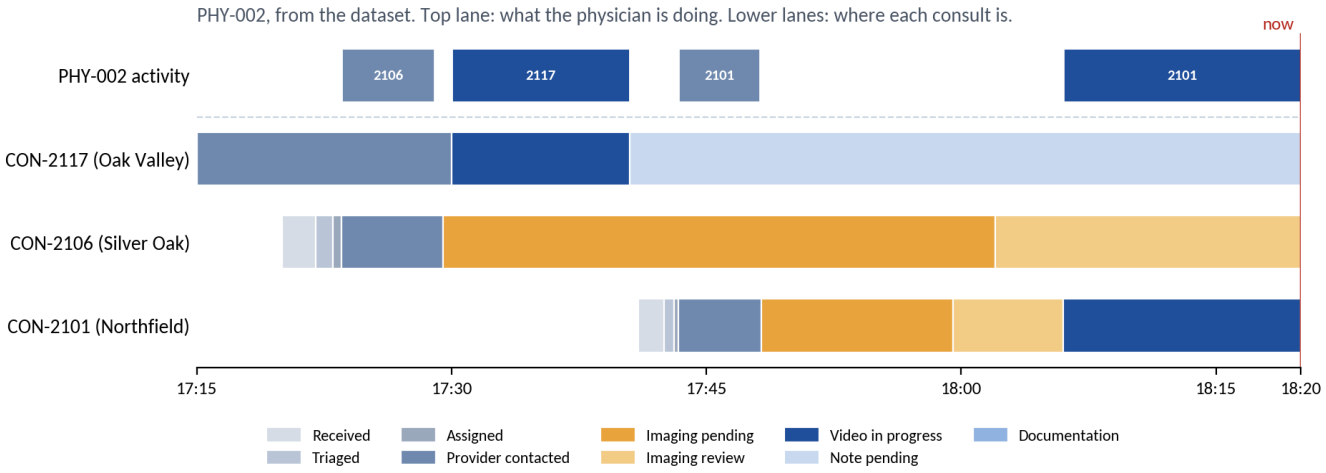


Figure 3. Built from the event stream. PHY-002 moves between calls and video while three consults progress independently. At 18:20 PHY-002 is on video for CON-2101 while CON-2106 waits in imaging review and CON-2117's note has not been started.

Availability is inferred, not reported

Physicians do not manually report that they are on a call or on video. Activity is derived from system signals, and those signals can disagree. Your solution must reconcile them.

Signal	Source	Values
Self-reported status	Physician sets it in Synapse	Available, In Meeting, Temporarily Unavailable

Signal	Source	Values
Hospital call	Synapse telephony events	Incoming or outgoing call active / ended
Synapse video session	Synapse video events	Video started / ended
Camera activity	Desktop presence agent on the physician's computer	Camera active / inactive
Agent heartbeat	Desktop presence agent	Online / offline (offline means camera state is unknown)

- On a Synapse call or video session: **busy**.
- Camera active with no Synapse video session and status not In Meeting: probably on an external video platform. Treat as **busy** and flag it.
- Camera active with status In Meeting: **available** (internal meetings are interruptible).
- Agent offline: routing assumes the physician is **available**, but this is unverified. Surface the risk.
- Events can arrive late. Every event carries both `event_time` and `ingested_at`. State built only from arrival order will be wrong.

Consult lifecycle

Received → Triaged → Assigned → Provider Contacted → CT/CTA or Imaging Pending → Imaging Review → Video Consult In Progress → Note Pending → Documentation → Signed Off. Not every consult passes through every stage. Phone-only consults skip video. Note Pending means the clinical interaction is over but documentation has not started; Documentation means the note is being written.

How Calls Reach Physicians (Synthetic Routing Rules)

You need these rules to predict whether an intervention actually relieves a bottleneck. Thresholds are synthetic.

1. Eligible physicians are on shift **and** credentialed at the calling hospital.
2. Remove anyone busy (see availability rules), anyone whose phone is already ringing for another call, and anyone who ended a call less than 2 minutes ago.
3. Physicians with low volume this shift (6 or fewer consults on days, 4 or fewer on nights) get priority at the hospitals they are credentialed for.
4. Otherwise, route to the eligible physician with the **fewest consults this shift**. Ties go to whoever answered least recently.
5. If nobody is eligible, the routing attempt fails and retries every 5 minutes. In the event stream this appears as `routing_attempt_failed`.

> **The split-shift trap.** A physician who joins mid-shift starts with zero consults. Under rule 4, routing will send them nearly every call at every hospital they are credentialed for, not just the hospitals that needed help. The new physician gets overwhelmed and the original imbalance moves rather than resolves. Your solution must address this, for example by seeding a starting consult count, limiting the joining physician to the gap hospitals, or another approach you can justify.

Operational Standards (Synthetic)

Stage	Applies to	Warning	Critical
Triaged, not yet assigned	All	5 min	10 min
Assigned, physician not yet contacted	Stroke	3 min	8 min
CT/CTA or Imaging Pending	Stroke	12 min	20 min
Imaging Review (imaging back, video not started)	Stroke	6 min	12 min
Note Pending + Documentation, from end of video or call	Stroke	30 min	45 min
Note Pending + Documentation, from end of video or call	General Neuro	45 min	90 min
Open notes within 60 min of the physician's shift end	All	Any	2 or more

Coverage and Split-Shift Eligibility

Coverage gap. Each hospital has a minimum number of available, credentialed physicians per time block (`coverage_requirements.csv`). A gap exists when that number is not met, either now or at a projected time (for example, when shifts end at 19:00). A demand spike (trailing 60-minute consults above the hospital's p90 baseline for that hour) raises the requirement by one.

An off-shift physician may receive a split-shift offer only if all of the following hold:

- They have opted in to split-shift offers.
- They are credentialed at one or more of the hospitals with the gap. Being credentialed elsewhere does not help.
- At least 10 hours have passed since their last shift ended, measured at the offered start time.
- Hours worked in the trailing 7 days plus the offered hours do not exceed 60.
- The offer does not overlap a shift they are already scheduled for.
- Offers are at least 2 hours long. The estimated time to come online counts against the value of the offer.

Notification and Safety Constraints

- No patient information in any notification. Hospital names, hours and start time only.
- Every offer is approved by the clinical lead on duty before it is sent.
- No more than one open offer per physician, and no more than 3 offers per physician per week.
- Handle the case where two physicians accept the same offer.
- Offers expire if not answered within 10 minutes.
- Log every offer: who was considered, who was excluded and why, who approved, and the outcome.

Synthetic Dataset

All files are in the accompanying data package. The scenario covers Tuesday 17 March 2026, 07:00-18:20 Eastern. **The reference point for "now" is 18:20:00 ET.** All timestamps include an explicit UTC offset; hospitals are in four US time zones.

File	Contents
event_stream.csv	About 1,160 events from 07:00 to 18:20. The source of truth. Sorted by ingested_at.
hospitals.csv	12 hospitals with local time zone and coverage tier
physicians_on_shift.csv	8 physicians currently on shift, with shift times
physicians_off_shift_roster.csv	6 off-shift physicians: opt-in, last and next shift, hours worked, time to come online
credentialing_matrix.csv	Which of the 14 physicians may take consults from which of the 12 hospitals
baseline_hourly_volume.csv	12-week weekday mean and p90 consults per hospital per hour
coverage_requirements.csv	Minimum available credentialed physicians per hospital per time block
snapshot_*_1820.csv	Reference state at 18:20. Your replay of the event stream should reproduce it.

Event types

`consult_received, consult_triaged, physician_assigned, incoming_call_ringing, incoming_call_started/ended, outgoing_call_started/ended, imaging_requested, imaging_returned, video_started/ended, note_started, note_signed, routing_attempt_failed, physician_status_changed, camera_active/inactive, agent_heartbeat_lost, shift_started`

Reference snapshot: physicians at 18:20 ET

ID	Physician	Shift (ET)	Self-reported	Agent	Derived state	Cons.	Calls	Open	Min to end
PHY-001	Dr. Maya Patel	08:00-20:00	Available	Online	Camera Active - No Synapse Session	12	13	1	100
PHY-002	Dr. Daniel Kim	08:00-20:00	Available	Online	Video Consult In Progress	14	16	3	100
PHY-003	Dr. Sofia Ramirez	07:00-19:00	Available	Online	Outgoing Hospital Call	14	17	3	40
PHY-004	Dr. Ethan Brooks	09:00-21:00	Available	Online	Documentation / Note	14	18	2	160
PHY-005	Dr. Priya Shah	08:00-20:00	Available	Online	Incoming Hospital Call	15	18	2	100
PHY-006	Dr. Marcus Lee	07:00-19:00	Available	Online	Video Consult In Progress	15	19	3	40
PHY-007	Dr. Elena Torres	10:00-22:00	Available	Offline	Available	6	6	0	220
PHY-008	Dr. Noah Williams	08:00-20:00	Temporarily Unavailable	Online	Temporarily Unavailable	10	11	1	100

The self-reported status reads Available for most physicians who are busy. That is expected: activity is derived, not reported.

Reference snapshot: open consults at 18:20 ET

Consult	Hospital	Physician	Type	Current stage	Entered	Min in stage	Min since received
CON-2101	Northfield Medical (SYN-008)	PHY-002	Stroke	Video Consult In Progress	18:06:00	14	39
CON-2102	Lakeview Hospital (SYN-009)	PHY-003	Stroke	CT/CTA or Imaging Pending	17:54:00	26	36
CON-2103	Cedar Ridge Medical Center (SYN-001)	PHY-004	General Neuro	Note Pending	17:46:00	34	75
CON-2104	Meadowlands Community Hospital (SYN-004)	PHY-005	Stroke	Provider Contacted	18:14:00	6	9
CON-2105	Oak Valley Medical (SYN-010)	Unassigned	Stroke	Triaged	18:09:05	10	12
CON-2106	Silver Oak Health Center (SYN-007)	PHY-002	General Neuro	Imaging Review	18:02:00	18	60
CON-2107	Suncrest Medical Pavilion (SYN-005)	PHY-006	Stroke	Video Consult In Progress	18:11:20	8	15
CON-2108	Redwood University Medical Center (SYN-003)	PHY-004	Stroke	Documentation	18:04:00	16	33
CON-2109	Westbridge Hospital (SYN-012)	Unassigned	General Neuro	Received	18:16:00	4	4
CON-2110	Harbor Regional (SYN-011)	PHY-008	Stroke	Assigned	18:07:15	12	16
CON-2111	Suncrest Medical Pavilion (SYN-005)	PHY-003	Stroke	Documentation	18:00:10	19	39
CON-2112	Suncrest Medical Pavilion (SYN-005)	PHY-006	Stroke	Imaging Review	18:04:50	15	29
CON-2113	Suncrest Medical Pavilion (SYN-005)	PHY-005	Stroke	Note Pending	18:11:50	8	25
CON-2114	Suncrest Medical Pavilion (SYN-005)	Unassigned	Stroke	Triaged	18:17:10	2	4
CON-2115	Lakeview Hospital (SYN-009)	PHY-003	General Neuro	Note Pending	17:30:00	50	90
CON-2116	Highland Valley Hospital (SYN-006)	PHY-001	General Neuro	Note Pending	17:58:00	22	70
CON-2117	Oak Valley Medical (SYN-010)	PHY-002	General Neuro	Note Pending	17:40:30	39	99
CON-2118	Westbridge Hospital (SYN-012)	PHY-006	General Neuro	Note Pending	17:35:00	45	78

Event stream excerpt (ingested 18:09-18:20 ET)

Event	Event time	Ingested	Consult	Type	Phys.	Hosp.	Description
EVT-01143	18:09:05	18:09:07	CON-2105	consult_triaged		SYN-010	Consult triaged
EVT-01144	18:09:10	18:09:13	CON-2105	routing_attempt_failed		SYN-010	No available credentialed provider; retry scheduled
EVT-01145	18:10:40	18:10:41	CON-2107	incoming_call_ended	PHY-006	SYN-005	Call ends
EVT-01146	18:10:45	18:10:46		physician_status_changed	PHY-006		Self-reported status: Available
EVT-01147	18:11:00	18:11:03	CON-2104	consult_received		SYN-004	Stroke consult request received
EVT-01148	18:11:22	18:11:23		camera_active	PHY-006		Desktop agent: camera active
EVT-01149	18:11:50	18:11:53	CON-2113	video_ended	PHY-005	SYN-005	Video consult completed; note not started
EVT-01150	18:11:53	18:11:54		camera_inactive	PHY-005		Desktop agent: camera inactive
EVT-01151	18:12:30	18:12:33	CON-2104	consult_triaged		SYN-004	Consult triaged
EVT-01152	18:12:40	18:12:41		physician_status_changed	PHY-008		Self-reported status: Temporarily Unavailable
EVT-01153	18:13:40	18:13:41	CON-2104	physician_assigned	PHY-005	SYN-004	Consult assigned
EVT-01154	18:14:00	18:14:03	CON-2104	incoming_call_started	PHY-005	SYN-004	Hospital calls assigned physician
EVT-01155	18:14:10	18:14:12	CON-2105	routing_attempt_failed		SYN-010	No available credentialed provider; retry scheduled
EVT-01156	18:15:30	18:15:33	CON-2114	consult_received		SYN-005	Stroke consult request received
EVT-01157	18:16:00	18:16:03	CON-2109	consult_received		SYN-012	General Neuro consult request received
EVT-01158	18:16:20	18:16:21	CON-2102	outgoing_call_started	PHY-003	SYN-009	Provider calls hospital to check CT/CTA status
EVT-01159	18:17:10	18:17:14	CON-2114	consult_triaged		SYN-005	Consult triaged
EVT-01160	18:17:15	18:17:18	CON-2114	routing_attempt_failed		SYN-005	No available credentialed provider; retry scheduled
EVT-01161	18:11:20	18:18:05	CON-2107	video_started	PHY-006	SYN-005	Video consult begins
EVT-01162	18:19:10	18:19:14	CON-2105	routing_attempt_failed		SYN-010	No available credentialed provider; retry scheduled
EVT-01163	18:19:40	18:19:43	CON-2109	incoming_call_ringing	PHY-007	SYN-012	Hospital call ringing to intended physician; not yet answered

Credentialing matrix (Y = credentialed)

Physician	001	002	003	004	005	006	007	008	009	010	011	012
PHY-001 Dr. Maya Patel	Y			Y		Y		Y	Y		Y	Y
PHY-002 Dr. Daniel Kim		Y					Y	Y		Y	Y	
PHY-003 Dr. Sofia Ramirez	Y			Y	Y				Y			Y
PHY-004 Dr. Ethan Brooks	Y		Y			Y	Y		Y			
PHY-005 Dr. Priya Shah			Y	Y	Y			Y			Y	
PHY-006 Dr. Marcus Lee	Y			Y	Y	Y		Y	Y		Y	Y
PHY-007 Dr. Elena Torres		Y										Y
PHY-008 Dr. Noah Williams						Y	Y			Y	Y	
PHY-009 Dr. Grace Okafor				Y	Y			Y		Y		
PHY-010 Dr. Ben Alvarez		Y	Y			Y	Y					
PHY-011 Dr. Hannah Cho	Y			Y	Y			Y	Y		Y	
PHY-012 Dr. Omar Haddad		Y	Y			Y	Y			Y		Y
PHY-013 Dr. Lucy Brennan					Y				Y	Y	Y	
PHY-014 Dr. Ravi Menon	Y				Y					Y		

Column headers are hospital IDs without the SYN- prefix. PHY-001 to PHY-008 are on shift; PHY-009 to PHY-014 are off shift.

Off-shift roster

ID	Physician	Role	Opt-in	Last shift ended	Next shift starts	Hrs 7d	Min to online	Channel	Declined 30d
PHY-009	Dr. Grace Okafor	Day	Yes	03-17 07:00	03-18 19:00	36	15	app_push	1
PHY-010	Dr. Ben Alvarez	Float	Yes	03-16 20:00	03-19 08:00	24	10	sms	0
PHY-011	Dr. Hannah Cho	Night	Yes	03-17 07:00	03-17 19:00	48	0	app_push	2
PHY-012	Dr. Omar Haddad	Night	No	03-16 08:00	03-17 20:00	36	0	app_push	0
PHY-013	Dr. Lucy Brennan	Day	No	03-16 20:00	03-18 08:00	48	20	sms	5
PHY-014	Dr. Ravi Menon	Float	Yes	03-16 20:00	03-19 07:00	59	40	sms	0

PHY-011 and PHY-012 are night physicians whose shifts start at 19:00 and 20:00. A scheduled night shift is not a split-shift offer.

Operational Situations Hidden in the Dataset

The dataset contains every situation below. Some are visible in the snapshot; others only appear when you replay the event stream correctly or project forward in time.

- A stroke consult is triaged but unassigned because every credentialed physician is busy or unavailable.
- A hospital generates several consults in under an hour, well above its baseline.
- Two physicians are 40 minutes from the end of their shift with several open consults, including unfinished notes, during that surge.
- A physician becomes temporarily unavailable while a stroke consult is still assigned to them.
- A physician's camera is active with no Synapse session.
- A physician looks available, but their presence agent is offline.
- A video-start event arrives several minutes after it happened.
- One event appears twice in the stream.
- Workload is uneven, and the cause is credentialing rather than effort.
- An incoming call rings while most on-shift physicians are occupied.
- Exactly one off-shift physician is fully eligible to cover the projected gap right now. Several others look eligible but fail one rule each.

What Teams Must Demonstrate

1. Rebuild current state from the event stream, handling late and duplicate events, and match the reference snapshot.
2. Show physician activity and the consult lifecycle independently, with time in stage measured against the standards.
3. Detect and rank what needs attention, and explain why each item was flagged.

4. Project coverage forward at least 90 minutes, accounting for shift ends and the demand baseline.
5. Recommend an intervention for the top issue. For a split shift, show every off-shift physician considered and the rule each one passed or failed.
6. Run the full loop: propose, approve, send (simulated), accept or decline, update projected coverage.
7. Show how you handle the split-shift trap.
8. **Live test.** During your demo, judges will add new events in the same format (for example, another stroke consult at the surging hospital). Your system must update without a restart.

Out of Scope

Clinical decisions about individual patients; real scheduling, credentialing or messaging systems; compensation for split shifts; building the production routing engine (model its effect, do not replace it).

Data and Safety Boundary

All physicians, hospitals, consults, timestamps, activity states, credentials, schedules and events in this challenge are synthetic. Thresholds and routing rules are illustrative and do not reflect Sevaro's production configuration. No patient identifiers, PHI, real physician activity, hospital production data, credentials or production-system access will be provided or required.